

A7 10 Inferencia

1) $X_1, \dots, X_n \sim \text{Cauchy}(\theta)$, $\theta \in \mathbb{R}$ obs: $E(X)$ no existe

$$f(x|\theta) = \frac{1}{\pi(1+(x-\theta)^2)}, x \in \mathbb{R}$$

a) $L(X|\theta) = \prod_{i=1}^n \frac{1}{\pi(1+(x_i-\theta)^2)}$

$$\ell(\theta) = \sum_{i=1}^n -\log(\pi(1+(x_i-\theta)^2)) = -\sum_{i=1}^n \log(1+(x_i-\theta)^2) - n \log(\pi)$$

b) $S(\theta) = \frac{\partial \ell}{\partial \theta} = -\sum_{i=1}^n \frac{2(x_i-\theta)}{1+(x_i-\theta)^2} = 2 \sum_{i=1}^n \frac{x_i-\theta}{1+(x_i-\theta)^2} \rightarrow \frac{\partial S}{\partial \theta} = \frac{2 \sum_{i=1}^n -(1+(x_i-\theta)^2) + 2(x_i-\theta)^2}{[1+(x_i-\theta)^2]^2}$

c) $\theta^{k+1} = \theta^k - \frac{S(\theta)}{S'(\theta)} = \theta^{(k)} + \frac{S(\theta)}{I(\theta)}$

Newton-

$$\theta^{k+1} = \theta^{(k)} + \frac{\sum_{i=1}^n \frac{(x_i-\theta)}{1+(x_i-\theta)^2}}{\sum_{i=1}^n \frac{1-(x_i-\theta)^2}{[1+(x_i-\theta)^2]^2}}$$

$$I(\theta) = -\frac{\partial S}{\partial \theta} = 2 \sum_{i=1}^n \frac{(x_i-\theta)^2 - 1}{[1+(x_i-\theta)^2]^2} = 2 \sum_{i=1}^n \frac{1-(x_i-\theta)^2}{[1+(x_i-\theta)^2]^2}$$

d) Dado $I_{X_1} = 1/2 \rightarrow I_{\bar{X}} = n/2$

Fisher Score

$$\theta^{(k+1)} = \theta^{(k)} + \frac{S(\theta)}{\frac{1}{n} I(\theta)} = \theta^{(k)} + \frac{2 \sum_{i=1}^n \frac{(x_i-\theta)}{1+(x_i-\theta)^2}}{\frac{n}{2}}$$

e) Ver código

P2 | $Y_i \sim \text{Poisson}(\lambda_i = \exp(a + bx_i))$, $a + bx_i > 0$
 x_i correlated,

$$P(Y_i = k | \lambda_i) = \frac{\lambda_i^k e^{-\lambda_i}}{k!} \quad k=0,1,\dots$$

$$\lambda_i = \exp(a + bx_i) \quad \forall i=1, \dots, n$$

$$a) \quad L(\vec{Y} | \hat{\lambda}, \vec{x}) = \frac{\prod_{i=1}^n \lambda_i^{y_i} e^{-\lambda_i}}{\prod_{i=1}^n y_i!} = \frac{\prod_{i=1}^n \lambda_i^{y_i} e^{-\sum_{i=1}^n \lambda_i}}{\prod_{i=1}^n y_i!}$$

$$l(\lambda) = \sum_{i=1}^n y_i \log(\lambda_i) - \sum_{i=1}^n \lambda_i - \sum_{i=1}^n y_i! \quad , \quad \lambda_i = \exp(a + bx_i)$$

$$= \sum_{i=1}^n y_i (a + bx_i) - \sum_{i=1}^n e^{a + bx_i} - C$$

$$l(a, b) = \sum_{i=1}^n y_i (a + bx_i) - e^{a + bx_i} - C \quad / \frac{\partial}{\partial a}$$

$$S(a, b)_1 = \frac{\partial l}{\partial a} = \sum_{i=1}^n y_i - e^{a + bx_i}$$

$$\frac{\partial l}{\partial b} = \sum_{i=1}^n y_i x_i - x_i e^{a + bx_i}$$

$$\frac{\partial l}{\partial b} = \sum_{i=1}^n x_i (y_i - e^{a + bx_i})$$

$$\frac{\partial^2 l}{\partial a^2} = - \sum_{i=1}^n e^{a + bx_i} \quad , \quad \frac{\partial^2 l}{\partial b^2} = - \sum_{i=1}^n x_i^2 e^{a + bx_i}$$

$$\frac{\partial^2 l}{\partial a \partial b} = - \sum_{i=1}^n x_i e^{a + bx_i}$$

$$c) \quad I(a, b) = - \frac{\partial S}{\partial(a, b)} = \begin{bmatrix} \sum_{i=1}^n e^{a + bx_i} & \sum_{i=1}^n x_i e^{a + bx_i} \\ \sum_{i=1}^n x_i e^{a + bx_i} & \sum_{i=1}^n x_i^2 e^{a + bx_i} \end{bmatrix}$$

$$= \sum_{i=1}^n \begin{bmatrix} \lambda_i & x_i \lambda_i \\ x_i \lambda_i & x_i^2 \lambda_i \end{bmatrix}$$

d) Newton: $\theta^{(k+1)} = \theta^{(k)} + \frac{J(\theta)^{-1} S(\theta)}{J(\theta)}$

$$E(I(\theta)) = I(\theta)$$

e) Ver código y comparar con la función glm.

P3] $X_1, \dots, X_n \sim \text{Exp}(\theta), \theta > 0$

~~Y~~

$$f(x|\theta) = \theta e^{-x\theta}$$

$$f_Y(y|\theta) = \frac{\theta}{\sigma} e^{-y} = e^{-y}$$

$$X_1 \theta = Y_1$$

$$X_1 \theta = \frac{Y_1}{\sigma} \quad \left| \frac{\partial}{\partial Y_1} \right.$$

$$\frac{\partial X_1}{\partial Y_1} = \frac{1}{\sigma}$$

$$Y \sim \text{exp}(1)$$

$$2Y = Z \quad f_Z(z) = \frac{1}{2} e^{-z/2}$$

$$Y = \frac{Z}{2}$$

$$Z \sim \text{exp}(1/2)$$

~~$\frac{1}{2} f_Y(y) = \frac{1}{2} e^{-y} = \frac{1}{2} e^{-z/2}$~~
 $\frac{1}{2} f_Y(y) = \frac{1}{2} e^{-y} = \frac{1}{2} e^{-z/2}$

Estadística suficiente $\sum X_i \sim \text{Gamma}(n, \theta)$

$$\theta \sum X_i = \sum Y_i \sim \text{Gamma}(n, 1)$$

$$2 \sum Y_i \sim \text{Gamma}(n, 1/2) = \text{Gamma}\left(\frac{2 \cdot n}{2}, 1/2\right) = \chi^2_{2n}$$

$$2\theta \sum X_i \sim \chi^2_{2n}$$

$$P_\theta(a \leq 2\theta \sum X_i \leq b) = 1 - \alpha$$

$$P_\theta\left(\chi^2_{2n}(\alpha/2) \leq 2\theta \sum X_i \leq \chi^2_{2n}(1 - \alpha/2)\right) = 1 - \alpha$$

$$\frac{\chi^2_{2n}(\alpha/2)}{2 \sum X_i} \leq \theta \leq \frac{\chi^2_{2n}(1 - \alpha/2)}{2 \sum X_i}$$

P4] Soit $X_1, \dots, X_n$ indépendantes, $X_i \sim N(\theta x_i, \sigma^2)$ $x_1, \dots, x_n$ connus
 $\sigma^2 > 0$ connu.

a) EMV de σ .

$$f(X_i | \theta, x_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2} (y_i - \theta x_i)^2\right\}$$

$$\ell f(y_i) = \log(\sqrt{2\pi}\sigma) - \frac{1}{2\sigma^2} (y_i - \theta x_i)^2$$

$$\ell f(\vec{y}) = \sum_{i=1}^n \left(\log(\sqrt{2\pi}\sigma) - \frac{1}{2\sigma^2} (y_i - \theta x_i)^2 \right) \quad / \quad \frac{\partial}{\partial \sigma}$$

$$\frac{\partial \ell f(\vec{y})}{\partial \sigma} = -\frac{1}{2\sigma^2} \sum_{i=1}^n x_i (y_i - \theta x_i) \cdot 2$$

$$= \frac{1}{\sigma^2} \sum_{i=1}^n x_i y_i - \theta \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n x_i y_i - \theta \sum_{i=1}^n x_i^2 = 0$$

$$\boxed{\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \hat{\theta}_{EMV}}$$

Conditionnelle linéaire de
Normal et
Normal

$$E(\hat{\theta}_{EMV}) = \frac{\sum_{i=1}^n x_i E(y_i)}{\sum_{i=1}^n x_i^2} = \sigma \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2} = \sigma$$

$$Var(\hat{\theta}_{EMV}) = \frac{\sum_{i=1}^n x_i^2 Var(y_i)}{(\sum_{i=1}^n x_i^2)^2} = \frac{\sigma^2}{\sum_{i=1}^n x_i^2} \sim N(0,1)$$

$$\hat{\theta}_{EMV} \sim N\left(\sigma, \frac{\sigma^2}{\sum_{i=1}^n x_i^2}\right) \quad P\left(a \leq \frac{\hat{\theta}_{EMV} - \sigma}{\sigma / \sqrt{\sum_{i=1}^n x_i^2}} \leq b\right) = 1 - 2$$

$$a = \frac{z_{\alpha/2}}{\alpha/2}$$

$$b = z_{1-\alpha/2}$$

$$z_{-\alpha/2} = -z_{1-\alpha/2}$$

$$z_{\alpha/2} \leq \frac{\hat{\theta}_{EMV} - \theta}{\sigma / \sqrt{\sum x_i^2}} \leq z_{1-\alpha/2}$$

$$z_{\alpha/2} \frac{\sigma}{\sqrt{\sum x_i^2}} \leq \hat{\theta}_{EMV} - \theta \leq \frac{\sigma}{\sqrt{\sum x_i^2}} z_{1-\alpha/2} \quad | - \hat{\theta}_{EMV}$$

$$- \frac{\sigma}{\sqrt{\sum x_i^2}} z_{1-\alpha/2} + \hat{\theta}_{EMV} \leq \theta \leq \frac{\sigma}{\sqrt{\sum x_i^2}} z_{1-\alpha/2} + \hat{\theta}_{EMV}$$

$$\hat{\theta}_{EMV} - \frac{\sigma}{\sqrt{\sum x_i^2}} z_{1-\alpha/2} \leq \theta \leq \hat{\theta}_{EMV} + \frac{\sigma}{\sqrt{\sum x_i^2}} z_{1-\alpha/2}$$

$$\theta \in \hat{\theta}_{EMV} \pm \frac{\sigma}{\sqrt{\sum x_i^2}} z_{1-\alpha/2}$$